Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/22/2005
PI Name:	Wold, Barbara Ph.D.		
Project Title:	Discovery and Definition of the Microgravity Transcriptome, with Application to Muscle Atrophy and Regeneration Processes		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Musculoskeletal Biology		
Space Biology Special Category:	None		
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Zip Code:	91125	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2000 Space Biology 00-OBPR-01
Start Date:	03/01/2002	**End Date:**	05/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NCE to 5/31/2006, per A. Chu 7/2006		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Shapiro, Bruce (NASA JPL)		
Grant/Contract No.:	NAG2-1540		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The project has integrated goals in development and application of bioinformatic tools for storing, analyzing and mining large scale gene expression data and comparative genomic data. Once primary data analysis is complete, it is integrated with andditional public genomic, proteomic, and functional annotation data. The informatics tools are thus applied to wet-bench microarray measurements using material from microgravity models. To develop bioinformatics tools that will be publicly available for use by the community. To use these and additional tools to discover the cellular pathways and circuits that control muscle atrophy, with emphasis on microgravity paradigms. Alterations in gravity impact cells in many ways, including cell proliferation, chromosomal aberations, gene expression, and processes of reproductive cell formation. At the systemic level, microgravity affects virtually every system in the body. Known detremental changes include bone mineral density losses that vary across specific bone sites with greater loss in the weight-bearing bones. Marked atrophy of postural muscles as well as reduced muscle strength and tasks prevent astronauts from performing sustained work in space and cause severe side-effects upon return to Earth. These effects increase with time in flight and it is not known how long they continue to develop. This project will help to understand the molecular and cellular causes of muscle atrophy and identify potential targets for counteremasure.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Earth benefits from this work are in three areas: 1) The bioinformatics and analysis tools are applicable to all microarray data including vast array of medical analysis and diagnostics using genomic methods. 2)Microgravity models and medical problems of muscle and bone degeneration asociated with, for example, extended bedrest, make the identification of genes involved in these pathways of significance to a large population of patients confined to bedrest.
Task Progress:	Rat hindlimb suspension samples: Skeletal muscle samples of soleus, gastrocnemius and tibialis anterior from rats on a one week hindlimb suspension protocol were processed for total RNA extraction, along with samples of left cardiac ventricle. Total RNA yield per milligram of tissue mass was significantly reduced for soleus and gastrocnemius compared to ambulatory control rats, confirming previously published results. Total RNA samples were reverse transcribed into double stranded cDNA containing a T7 promoter site, from which biotin labeled cRNA was produced. cRNA samples were analyzed for consistent yield and length distribution by electropherogram, prior to hybridization to Affymetrix Rat Expression set 230 version 2.0 chips. Each chip assays 30,248 expressed sequences and variants using 11 probe pairs per sequence. The average portion of chip capacity reliably detected in the eight hybridizations performed was 46% ± 4%. In the initial expression analysis, a singular value decomposition method was used to identify eight principal components (PCs) for the eight factor dataset comprised of genes that were reliably detected in both experimental and control soleus muscle, which has previously been shown to be most sensitive to hindlimb suspension. Contributions of each gene vector to each PC were fit to a Gaussian distribution, and ranked according to variance from the PC mean. The 100 genes with largest variance were identified as a significant “outlier” group characteristic of each PC. Plots of the contribution to each PC from each of the 8 tissue samples identified a PC that captures the variance that distinguishes cardiac muscle from skeletal muscle, and a PC that captures the variance that distinguishes soleus and gastrocnemius from tibialis anterior and left cardiac ventricle. The 100 outlier genes from each of the 8 PCs will be analyzed for statistically significant over- or under-representation of specific Gene Ontology (GO) categories. The GO categories represented are being used as one guide to objectively highlight the cellular and molecular pathways and circuits implicated in the in the atrophy process in vivo. Additional analyses integrate these new data with global proteomic data and with expression pathway information from studies of RNA expression over developmental and regeneration regimes performed outside this study. This information can now be used, in conjunction with existing knowledge of major muscle signaling pathways to generate a prototype mathematical model using Cellerator. We have also completed printing over 900 microarray chips with a collection of 70mer oligonucleotides representing 32,829 mouse transcripts covering 24,878 genes from the Ensembl Mouse 1430 database, and are currently testing them for performance by cohybridization of labeled cDNA and labeled mouse genomic DNA. Cohybridization with genomic DNA will reduce the variation in measurements due to unavoidable inconsistencies in the amount of DNA probe mass spotted on the array features. Our genomic DNA labeling protocol has been simplified for ease of use and reduction of labeling cost by 50%. The large number of arrays available will allow more experimental replicates, and a means for confirming and comparing the results obtained from the initial Affymetrix arrays. In addition, we have completed a detailed comparative genomic analysis of candidate regulatory domains of the MuRF1 gene, a primary regulator of muscle atrophy. We have identified five major evolutionarily conserved non-coding domains in the region of MuRF1. These are high quality candidates for a role in homeostatic regulation of atrophy as cis regultory elements or by encoding small regulatory RNAs. Cell culture and transgenic animal assays will be used to identify which candidate regulatory domains are functional and to ultimately determine which are active in the hind-limb suspension paradigm. A similar anaysis of Mafbx, a second E3ubiquitin ligase with a causal role in global muscle degeneration.
Bibliography Type:	Description: (Last Updated: 01/09/2020)
Articles in Peer-reviewed Journals	Williams BA, Gwirtz RM, Wold BJ. "Genomic DNA as a cohybridization standard for mammalian microarray measurements." Nucleic Acids Res. 2004 Jun 9;32(10):e81. PubMed PMID: 15190126; PubMed Central https://www.ncbi.nlm.nih.gov/pmc/articles/PMC434457/ , Jun-2004
Articles in Peer-reviewed Journals	Hart CE, Sharenbroich L, Bornstein BJ, Trout D, King B, Mjolsness E, Wold BJ. "A mathematical and computational framework for quantitative comparison and integration of large-scale gene expression data." Nucleic Acids Res. 2005 May 10;33(8):2580-94. PubMed PMID: 15886390; PubMed Central PMCID: PMC1092273 , May-2005